FULL VULNERABILITY ASSESSMENT REPORT

Training example • Fictional Linux virtual machine • 18 October 2026

SIMULATED REPORT — NOT REAL SCAN OUTPUT
No virtual machine was scanned for this document. Host details, ports, scanner findings and evidence below are illustrative placeholders. Replace them with results from your authorized lab VM before presenting this as a completed assessment.

1. Scope and Executive Summary

The assessment scenario covers a fictional Linux VM on a private training network. The objective is to identify exposed services, review likely software weaknesses, and provide prioritized remediation. In this sample, the overall risk is **High** because an outdated web service and an unnecessary remote administration service could increase the chance of unauthorized access. Findings must be confirmed against actual Nmap and OpenVAS/Nessus output before any real-world risk decision is made.

2. Nmap — Open Ports and Services (sample table)

Port/Protocol	Example service	Version / observation	Initial concern
22/tcp	SSH	OpenSSH version to verify	Remote login exposed; restrict access and enforce key/MFA policy where supported
80/tcp	HTTP	Web server version to verify	Cleartext HTTP; redirect to HTTPS if a web service is required
443/tcp	HTTPS	TLS/web server version to verify	Review TLS settings, certificate and patch status

Example authorized-lab command: `nmap -sV -oN nmap_results.txt <VM_IP>`. Use only against the VM you have permission to assess. Nmap `-sV` attempts to identify service versions; confirm results on the host where possible.

3. Vulnerability Scanner Findings (illustrative only)

ID	Example finding	Severity	Validation needed
V-01	Potential outdated web server package	High (example)	Confirm exact product/version and whether an applicable CVE is present in scanner output
V-02	SSH reachable from broader network than required	Medium (example)	Check firewall/security-group rules and authorized admin access requirements
V-03	HTTP service available without HTTPS redirection	Low/Medium (context-dependent)	Check whether sensitive traffic can be served over HTTP and inspect redirect/TLS configuration

Scanner step: run OpenVAS/Greenbone or Nessus against the same authorized VM, then export the report with vulnerability name, severity/CVSS, affected service, evidence and remediation. The rows above are sample scenarios, not detected vulnerabilities.

4. CVE Impact Research (example approach)

No CVE can responsibly be tied to the fictional service until the real product name and exact version are known. After the scan, select one High/Critical result, record its CVE ID, affected versions, CVSS score and vendor-fixed version, and check the official NVD record. Explain in plain language the attack prerequisites, potential result (such as remote code execution or information disclosure), affected data/features, and any mitigating conditions. Do not claim a CVE is exploitable merely because a port is open.

Official lookup: NIST National Vulnerability Database (NVD). CVSS is a severity metric; real risk also depends on exposure, exploitability, asset importance and controls.

5. Prioritized Recommendations

Priority	Action	Reason / verification
P1 — Immediate	Patch or upgrade any confirmed High/Critical vulnerable internet- or network-reachable service to the vendor-fixed version; apply compensating isolation if patching must wait.	Reduces risk of severe compromise. Re-scan and verify the fixed version/advisory.
P2 — High	Restrict SSH to an admin subnet/VPN, disable password login where suitable, use strong key-based authentication and remove unused accounts.	Reduces remote attack surface. Verify firewall rules and test authorized admin access.
P3 — Medium	Disable unused services/ports and allow only required sources in host firewall or VM network rules.	Limits reachable services. Repeat Nmap scan to confirm closure.
P4 — Planned	Redirect HTTP to HTTPS where applicable; maintain valid certificates and secure TLS configuration.	Protects traffic in transit. Test redirects and TLS configuration.

6. Peer Presentation and Q&A;

Short presentation: “I reviewed the VM’s exposed services with Nmap and used a vulnerability scanner to identify software or configuration issues. I ranked remediation by severity, exploitability, exposure and business impact. The first action is to fix any confirmed critical/high issue, then reduce unnecessary network access, and finally retest to confirm the changes.”

Q1: Why fix High/Critical issues first?
A: They may enable serious outcomes such as remote code execution or unauthorized data access, especially when reachable and exploitable. Priority should still be confirmed using the affected asset and actual evidence.

Q2: How will you prove the fixes worked?
A: Apply the vendor-recommended fix, verify the installed version/configuration, repeat the Nmap and vulnerability scans, and compare the new report with the original. Confirm that the finding is resolved and required services still work.

Appendix — Evidence and references

• Attach the actual Nmap text/XML output and OpenVAS/Nessus exported report when available.
• Nmap service/version detection: official Nmap documentation.

- CVE and CVSS details: NIST NVD.
- OpenVAS/Greenbone documentation: official documentation.